

Ej. 4

$$\underline{A} \times \underline{B} \times \underline{C} = (\underline{A} \cdot \underline{C}) \underline{B} - (\underline{A} \cdot \underline{B}) \underline{C}$$

$$(\underline{A} \times \underline{B} \times \underline{C})_i = (\underline{A} \times \underline{D})_i = \epsilon_{ijk} A_j D_k$$

$$(\underline{D})_k = D_k = (\underline{B} \times \underline{C})_k = \epsilon_{kmn} B_m C_n$$

$$(\underline{A} \times \underline{B} \times \underline{C})_i = \epsilon_{ijk} A_j D_k = \epsilon_{ijk} \epsilon_{kmn} A_j B_m C_n$$

$$= \epsilon_{kij} \epsilon_{kmn} A_j B_m C_n$$

$$= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) A_j B_m C_n =$$

$$= \delta_{im} \delta_{jn} A_j B_m C_n - \delta_{in} \delta_{jm} A_j B_m C_n =$$

$$= A_n B_i C_n - A_m B_m C_i$$

$$(\underline{A} \times \underline{B} \times \underline{C})_i = u_i$$

$$(\underline{A} \times \underline{B} \times \underline{C})_i = A_n C_n B_i - A_m B_m C_i$$

$$= (\underline{A} \cdot \underline{C}) \underline{B}_i - (\underline{A} \cdot \underline{B}) \underline{C}_i$$

$$= [(\underline{A} \cdot \underline{C}) \underline{B} - (\underline{A} \cdot \underline{B}) \underline{C}]_i$$

⇓

$$\underline{A} \times \underline{B} \times \underline{C} = (\underline{A} \cdot \underline{C}) \underline{B} - (\underline{A} \cdot \underline{B}) \underline{C}$$

ERROR

$$u_i = \underline{u}$$

sin símbolo de vector